

2.6 Ordinary and Partial Differential Equations

Multiplying (4) by x^2 , we get exact equation $(ayx^2 - y^2x^2 - 2x)dx + x^{-1}(2y - a)dy = 0$

By the usual rule of solving an exact equation, its solution is

$$\int \{(ay - y^2)x^{-2} - 2x\} dx + \int x^{-1}(2y - a) dy = b$$

(Treating y as constant) (Integrating terms free from x)

$$\text{or } (ay - y^2) \times (-1/x) - x^2 = b \quad \text{or} \quad (y^2 - ax)/x - x^2 = b$$

$$\text{or } (y^2 - ax - x^3)/x = b, \text{ where } b \text{ is an arbitrary constant.} \quad \dots (5)$$

From (3) and (5), required solution is $(y^2 - ax - x^3)/x = \phi(z/y)$, ϕ being an arbitrary function

EXERCISE 2(B)

Solved the following differential equations:

1. $p - 2q = 3x^2 \sin(y + 2x).$

Ans. $x^2 \sin(y + 2x) - z = \phi(y + 2x)$

2. $p - q = z/(x + y).$

Ans. $x - (x + y) \log z = \phi(x + y)$

3. $xy^2p - y^3q + axz = 0.$

Ans. $\log z + (ax/3y^2) = \phi(xy)$

4. $(x^2 - y^2 - z^2)p + 2xyq = 2xz.$

Ans. $(x^2 + y^2 + z^2)/z = \phi(y/z)$

5. (a) $z(p - q) = z^2 + (x + y)^2.$

Ans. $e^{2y}[z^2 + (x + y)^2] = \phi(x + y)$

[Agra 2011, Meerut 2011; Delhi B.A. (Prog.) II, 2014]

(b) $z(p + q) = z^2 + (x - y)^2$

Ans. $e^{2y}[z^2 + (x - y)^2] = \phi(x - y)$

6. $p - 2q = 3x^2 \sin(y + 2x).$

Ans. $x^3 \sin(y + 2x) - z = \phi(y + 2x)$

7. $p - q = z/(x + y).$

Ans. $x - (x + y) \log z = \phi(x + y)$

8. $zp - zq = x + y.$

Ans. $2x(x + y) - z^2 = \phi(x + y)$

9. $xyp + y^2q + 2x^2 - xyz = 0.$

Ans. $x - \log |z - (2x/y)| = \phi(x/y)$

2.9 Type 3 based on rule III for solving

$$(dx)/P = (dy)/Q = (dz)/R. \dots (1)$$

Let P_1, Q_1 and R_1 be functions of x, y and z . Then, by a well-known principle of algebra, each fraction in (1) will be equal to

$$(P_1 dx + Q_1 dy + R_1 dz) / (P_1 P + Q_1 Q + R_1 R). \dots (2)$$

If $P_1 P + Q_1 Q + R_1 R = 0$, then we know that the numerator of (2) is also zero. This gives $P_1 dx + Q_1 dy + R_1 dz = 0$ which can be integrated to give $u_1(x, y, z) = c_1$. This method may be repeated to get another integral $u_2(x, y, z) = c_2$. P_1, Q_1, R_1 are called multipliers. As a special case, these can be constants also. Sometimes only one integral is possible by use of multipliers. In such cases second integral should be obtained by using rule I of Art. 2.5 or rule II of Art. 2.7 as the case may be.

2.10 Solved examples based on Art. 2.9

Ex.1. Solve $\{(b - c)/a\}yzp + \{(c - a)/b\}zxq = \{(a - b)/c\}xy.$

Sol. Given $\{(b - c)/a\}yzp + \{(c - a)/b\}zxq = \{(a - b)/c\}xy. \dots (1)$

The Lagrange's subsidiary equations of (1) are $\frac{a dx}{(b - c)yz} = \frac{b dy}{(c - a)zx} = \frac{c dz}{(a - b)xy}. \dots (2)$

Choosing x, y, z as multipliers, each fraction for (2)

$$= \frac{axdx + bydy + czdz}{xyz[(b - c) + (c - a) + (a - b)]} = \frac{axdx + bydy + czdz}{0}.$$

$$\therefore ax dx + by dy + cz dz = 0 \quad \text{or} \quad 2axdx + 2bydy + 2czdz = 0.$$

Integrating, $ax^2 + by^2 + cz^2 = c_1$, c_1 being an arbitrary constant. ... (3)

Again, choosing ax , by , cz as multipliers, each fraction of (2)

$$= \frac{a^2 x dx + b^2 y dy + c^2 z dz}{xyz[a(b-c) + b(c-a) + c(a-b)]} = \frac{a^2 x dx + b^2 y dy + c^2 z dz}{0}.$$

$$\therefore a^2 x dx + b^2 y dy + c^2 z dz = 0 \quad \text{or} \quad 2a^2 x dx + 2b^2 y dy + 2c^2 z dz = 0.$$

Integrating, $a^2 x^2 + b^2 y^2 + c^2 z^2 = c_2$, c_2 being an arbitrary constant. ... (4)

From (3) and (4), the required general solution is given by

$$\phi(ax^2 + by^2 + cz^2, a^2 x^2 + b^2 y^2 + c^2 z^2) = 0, \text{ where } \phi \text{ is an arbitrary function.}$$

Ex. 2. Solve $(mz - ny)p + (nx - lz)q = ly - mx$. [Patna 2003; Madras 2005; Rohilkhand 2008; Bhopal 2004; Meerut 2008, 10; Sagar 2002; Lucknow 2009; Kanpur 2015]

Sol. The Lagrange's auxiliary equations for the given equation are

$$\frac{dx}{mz - ny} = \frac{dy}{nx - lz} = \frac{dz}{ly - mx}. \quad \dots (1)$$

Choosing x , y , z as multipliers, each fraction of (1)

$$= \frac{x dx + y dy + z dz}{x(mz - ny) + y(nx - lz) + z(ly - mx)} = \frac{x dx + y dy + z dz}{0}$$

$$\therefore x dx + y dy + z dz = 0 \quad \text{or} \quad 2x dx + 2y dy + 2z dz = 0$$

Integrating, $x^2 + y^2 + z^2 = c_1$, c_1 being an arbitrary constant. ... (2)

Again, choosing l , m , n as multipliers, each fraction of (1)

$$= \frac{l dx + m dy + n dz}{l(mx - ny) + m(nx - lz) + n(ly - mx)} = \frac{l dx + m dy + n dz}{0}.$$

$$\therefore l dx + m dy + n dz = 0 \quad \text{so that} \quad lx + my + nz = c_2. \quad \dots (3)$$

From (2) and (3), the required general solution is given by

$$\phi(x^2 + y^2 + z^2, lx + my + nz) = 0, \phi \text{ being an arbitrary function.}$$

Ex. 3. Solve $x(y^2 - z^2)q - y(z^2 + x^2)p = z(x^2 + y^2)$.

Sol. The Lagrange's auxiliary equations for the given equation are

$$\frac{dx}{x(y^2 - z^2)} = \frac{dy}{-y(z^2 + x^2)} = \frac{dz}{z(x^2 + y^2)}. \quad \dots (1)$$

Choosing x , y , z , as multipliers, each fraction of (1)

$$= \frac{x dx + y dy + z dz}{x^2(y^2 - z^2) - y^2(z^2 + x^2) + z^2(x^2 + y^2)} = \frac{x dx + y dy + z dz}{0}$$

$$\Rightarrow x dx + y dy + z dz = 0 \quad \text{so that} \quad x^2 + y^2 + z^2 = c_1. \quad \dots (2)$$

Choosing $1/x$, $-1/y$, $-1/z$ as multipliers, each fraction of (1)

$$= \frac{(1/x) dx - (1/y) dy - (1/z) dz}{y^2 - z^2 + z^2 + x^2 - (x^2 + y^2)} = \frac{(1/x) dx - (1/y) dy - (1/z) dz}{0}$$

$$\Rightarrow (1/x) dx - (1/y) dy - (1/z) dz = 0 \quad \text{so that} \quad \log x - \log y - \log z = \log c_2$$

$$\Rightarrow \log \{x/(yz)\} = \log c_2 \quad \Rightarrow \quad x/yz = c_2. \quad \dots (3)$$

$\therefore$ The required solution is $\phi(x^2 + y^2 + z^2, x/yz) = 0$, ϕ being an arbitrary function.

Ex. 4. Solve $(y - zx)p + (x + yz)q = x^2 + y^2$.

Sol. Try yourself.

Ans. $\phi(x^2 - y^2 + z^2, xy - z) = 0$

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Ex. 5. Solve $x(y^2 + z)p - y(x^2 + z)q = z(x^2 - y^2)$. [Kanpur 2013; Agra 2005; M.S. Univ. T.N. 2007; Indore 2003; Meerut 2009; Purvanchal 2007; Lucknow 2011]

Sol. Here Lagrange's subsidiary equations for given equation are

$$\frac{dx}{x(y^2 + z)} = \frac{dy}{-y(x^2 + z)} = \frac{dz}{z(x^2 - y^2)} \quad \dots(1)$$

Choosing $1/x, 1/y, 1/z$ as multipliers, each fraction of (1)

$$\begin{aligned} &= \frac{(1/x)dx + (1/y)dy + (1/z)dz}{y^2 + z - (x^2 + z) + x^2 - y^2} = \frac{(1/x)dx + (1/y)dy + (1/z)dz}{0} \\ \Rightarrow (1/x)dx + (1/y)dy + (1/z)dz &= 0 \quad \text{so that} \quad \log x + \log y + \log z = \log c_1 \\ \Rightarrow \log(xyz) &= \log c_1 \quad \Rightarrow \quad xyz = c_1. \quad \dots(2) \end{aligned}$$

Choosing $x, y, -1$ as multipliers, each fraction of (1)

$$\begin{aligned} &= \frac{xdx + ydy - dz}{x^2(y^2 + z) - y^2(x^2 + z) - z(x^2 - y^2)} = \frac{xdx + ydy - dz}{0} \\ \Rightarrow xdx + ydy - dz &= 0 \quad \text{so that} \quad x^2 + y^2 - 2z = c_2. \quad \dots(3) \end{aligned}$$

$\therefore$ From (2) and (3), solution is $\phi(x^2 + y^2 - 2z, xyz) = 0$, ϕ being an arbitrary function.

Ex. 6. Solve $(x + 2z)q + (4zx - y)q = 2x^2 + y$. [Meerut 2005]

Sol. Here Lagrange's auxiliary equations are $\frac{dx}{x+2z} = \frac{dy}{4zx-y} = \frac{dz}{2x^2+y}$. $\dots(1)$

Choosing $y, x, -2z$ as multipliers, each fraction of (1)

$$\begin{aligned} &= \frac{ydx + xdy - 2zdz}{y(x+2z) + x(4zx-y) - 2z(2x^2+y)} = \frac{d(xy) - 2zdz}{0} \\ \Rightarrow d(xy) - 2zdz &= 0 \quad \text{so that} \quad xy - z^2 = c_1. \quad \dots(2) \end{aligned}$$

Choosing $2x, -1, -1$ as multipliers, each fraction of (1)

$$\begin{aligned} &= \frac{2xdx - dy - dz}{2x(x+2z) - (4zx-y) - (2x^2+y)} = \frac{2xdx - dy - dz}{0} \\ \Rightarrow 2xdx - dy - dz &= 0 \quad \text{so that} \quad x^2 - y - z = c_2. \quad \dots(3) \end{aligned}$$

$\therefore$ From (2) and (3), solution is $\phi(xy - z^2, x^2 - y - z) = 0$, ϕ being an arbitrary function.

Ex. 7. Solve $(z^2 - 2yz - y^2)p + (xy + zx)q = xy - zx$. [Ranchi 2010; Lucknow 2007]

If the solution of the above equation represents a sphere, what will be the coordinates of its centre.

Sol. Here Lagrange's auxiliary equations for given equation are

$$\frac{dx}{z^2 - 2yz - y^2} = \frac{dy}{x(y+z)} = \frac{dz}{x(y-z)} \quad \dots(1)$$

Taking the last two fractions of (1), we have

$$(y-z)dy = (y+z)dz \quad \text{or} \quad 2ydy - 2zdz - 2(zdy + ydz) = 0. \quad \dots(2)$$

Integrating, $y^2 - z^2 - 2yz = c_1$, c_1 being an arbitrary constant.

Choosing x, y, z as multipliers, each fraction of (1)

$$\begin{aligned} &= \frac{xdx + ydy + zdz}{x(z^2 - 2yz - y^2) + xy(y+z) + xz(y-z)} = \frac{xdx + ydy + zdz}{0} \\ \Rightarrow 2xdx + 2ydy + 2zdz &= 0 \quad \text{so that} \quad x^2 + y^2 + z^2 = c_2. \quad \dots(3) \end{aligned}$$

From (2) and (3), solution is $\phi(y^2 - z^2 - 2yz, x^2 + y^2 + z^2) = 0$, ϕ being an arbitrary function.

From the solution of the given equation, it follows that if it represents a sphere, then its centre must be at $(0,0,0)$, i.e., origin.

Ex. 8. Solve $(y^3x - 2x^4)p + (2y^4 - x^3y)q = 9z(x^2 - y^3)$. [Jabalpur 2004; M.S. Univ. T.N. 2007]

Sol. Here Lagrange's auxiliary equations for the given equation are given by

$$\frac{dx}{y^3x - 2x^4} = \frac{dy}{2y^4 - x^3y} = \frac{dz}{9z(x^2 - y^3)} \quad \dots(1)$$

Taking first two fractions of (1), we have $(2y^4 - x^3y)dx = (y^3x - 2x^4)dy$

Dividing both sides by x^3y^3 gives $\left(\frac{2y}{x^3} - \frac{1}{y^2}\right)dx = \left(\frac{1}{x^2} - \frac{2x}{y^3}\right)dy$

$$\text{or} \quad \left(\frac{1}{x^2}dy - \frac{2y}{x^3}dx\right) + \left(\frac{1}{y^2}dx - \frac{2x}{y^3}dy\right) = 0 \quad \text{or} \quad d\left(\frac{y}{x^2}\right) + d\left(\frac{x}{y^2}\right) = 0.$$

Integrating, $(y/x^2) + (x/y^2) = c_1$, c_1 being an arbitrary constant. ...(2)

Choosing $1/x$, $1/y$, $1/3z$ as multipliers, each fraction of (1)

$$= \frac{(1/x)dx + (1/y)dy + (1/3z)dz}{(y^3 - 2x^3) + (2y^3 - x^3) + 3(x^3 - y^3)} = \frac{(1/x)dx + (1/y)dy + (1/3z)dz}{0}$$

$$\Rightarrow (1/x)dx + (1/y)dy + (1/3z)dz = 0 \quad \text{so that} \quad \log x + \log y + (1/3) \times \log z = \log c_2$$

$$\Rightarrow \log(xy z^{1/3}) = \log c_2 \quad \Rightarrow \quad xyz^{1/3} = c_2. \quad \dots(3)$$

From (2) and (3) solution is $\phi(xyz^{1/3}, y/x^2 + x/y^2) = 0$, ϕ being an arbitrary function.

Ex. 9. Solve $x^2p + y^2q = nxy$. [Jabalpur 2002]

Sol. Try yourself. **Ans.** $\phi\left(\frac{y-x}{xy}, z - \frac{nxy}{y-x} \log \frac{y}{x}\right) = 0$, ϕ being an arbitrary function.

Ex. 10. Solve $x(x^2 + 3y^2)p - y(3x^2 + y^2)q = 2z(y^2 - x^2)$. [Delhi B.Sc Hons 2000]

Sol. Here the Lagrange's auxiliary equations for the given equation are

$$\frac{dx}{x(x^2 + 3y^2)} = \frac{dy}{-y(3x^2 + y^2)} = \frac{dz}{2z(y^2 - x^2)} \quad \dots(1)$$

Choosing $1/x$, $1/y$, $-1/z$ as multipliers, each fraction of (1)

$$= \frac{(1/x)dx + (1/y)dy - (1/z)dz}{0} \quad \text{so that} \quad \frac{1}{x}dx + \frac{1}{y}dy - \frac{1}{z}dz = 0.$$

Integrating, $\log x + \log y - \log z = \log c_1$ so that $(xy)/z = c_1$(2)

$$\text{Taking the first two ratios of (1),} \quad \frac{dy}{dx} = -\frac{y(3x^2 + y^2)}{x(x^2 + 3y^2)} = -\left(\frac{y}{x}\right) \frac{3 + (y/x)^2}{1 + 3(y/x)^2} \quad \dots(3)$$

Put $y/x = v$ or $y = xv$ so that $(dy/dx) = v + x(dv/dx)$(4)

$$\text{Using (4), (3) reduces to} \quad v + x \frac{dv}{dx} = -v \frac{3 + v^2}{1 + 3v^2} \quad \text{or} \quad x \frac{dv}{dx} = -v \left[\frac{3 + v^2}{1 + 3v^2} + 1 \right]$$

$$\text{or} \quad x \frac{dv}{dx} = -\frac{4(1 + v^2)v}{1 + 3v^2} \quad \text{or} \quad 4 \frac{dx}{x} + \frac{1 + 3v^2}{v(1 + v^2)} dv = 0$$

$$\text{or} \quad 4 \frac{dx}{x} + \left(\frac{1}{v} + \frac{2v}{1 + v^2}\right) dv, \text{ on resolving into partial fractions}$$

Integrating, $4 \log x + \log v + \log(1 + v^2)$ or $x^4 v(1 + v^2) = c_2'$

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$x^4 (y/x)[1+(y/x)^2]=c'_2$ or $xy(x^2+y^2)=c'_2$ or $c_1z(x^2+y^2)=c'_2$, by (2)
 or $z(x^2+y^2)=c'_2/c_1$ or $z(x^2+y^2)=c_2$, where $c_2=c'_2/c_1$ (5)
 $\therefore$ From (2) and (5) solution is $\phi(z(x^2+y^2), xy/z)=0$, ϕ being an arbitrary function.

Ex. 11. Solve $(y-z)p + (z-x)q = x-y$. [Agra 2010; Kanpur 2017]

Sol. Here the Lagrange's auxiliary equations are $\frac{dx}{y-z} = \frac{dy}{z-x} = \frac{dz}{x-y}$ (1)

Choosing 1, 1, 1 as multipliers, each fraction of (1)

$$= \frac{dx+dy+dz}{(y-z)+(z-x)+(x-y)} = \frac{dx+dy+dz}{0}.$$

$\therefore dx+dy+dz=0$ so that $x+y+z=c_1$ (2)

Choosing x, y, z as multipliers, each fraction of (1)

$$= \frac{x dx + y dy + z dz}{x(y-z) + y(z-x) + z(x-y)} = \frac{x dx + y dy + z dz}{0}$$

$\therefore 2x dx + 2y dy + 2z dz = 0$ so that $x^2 + y^2 + z^2 = c_2$ (3)

$\therefore$ From (2) and (3) solution is $\phi(x+y+z, x^2+y^2+z^2)=0$, ϕ being an arbitrary of function.

Ex. 12. Solve the general solution of the equation $(y+zx)p - (x+yz)q + y^2 - x^2 = 0$.
[Meerut 2014; Delhi B.Sc. (Prog) II 2011; GATE 2001]

Sol. Given $(y+zx)p - (x+yz)q = x^2 - y^2$ (1)

Here the Lagrange's auxiliary equations are $\frac{dx}{y+zx} = \frac{dy}{-(x+yz)} = \frac{dz}{x^2-y^2}$ (2)

Choosing $x, y, -z$ as multipliers, each fraction of (2)

$$= \frac{x dx + y dy - z dz}{x(y+zx) - y(x+yz) - z(x^2-y^2)} = \frac{x dx + y dy - z dz}{0}$$

$\therefore x dx + y dy - z dz = 0$ so that $2x dx + 2y dy - 2z dz = 0$.
Integrating, $x^2 + y^2 - z^2 = c_1$, c_1 being an arbitrary constant. ... (3)

Choosing $y, x, 1$ as multipliers, each fraction of (2)

$$= \frac{y dx + x dy + dz}{y(y+zx) - x(x+yz) + x^2 - y^2} = \frac{y dx + x dy + dz}{0}$$

$\therefore y dx + x dy + dz = 0$ or $d(xy) + dz = 0$.
Integrating, $xy + z = c_2$, c_2 being an arbitrary constant. ... (4)

$\therefore$ The required solution is $\phi(x^2 + y^2 - z^2, xy + z) = 0$, ϕ being an arbitrary function.

Ex. 13. Solve $x(y-z)p + y(z-x)q = z(x-y)$, i.e., $\{(y-z)/(yz)\}p + \{(z-x)/(zx)\}q = (x-y)/(xy)$. [Delhi B.A (Prog) II 2010; Rohilkhand 2009, M.S. Univ. T.N. 2007; Meerut 2015]

Sol. Given $x(y-z)p + y(z-x)q = z(x-y)$... (1)

The Lagrange's auxiliary equations for (1) are $\frac{dx}{x(y-z)} = \frac{dy}{y(z-x)} = \frac{dz}{z(x-y)}$... (2)

Choosing $1/x, 1/y, 1/z$ as multipliers each fraction of (1)

$$= \frac{(1/x)dx + (1/y)dy + (1/z)dz}{(y-z)+(z-x)+(x-y)} = \frac{(1/x)dx + (1/y)dy + (1/z)dz}{0}$$

$\Rightarrow (1/x)dx + (1/y)dy + (1/z)dz = 0$ so that $\log x + \log y + \log z = \log c_1$
 $\therefore \log (xyz) = c_1$ or $xyz = c_1$... (3)

Choosing 1, 1, 1 as multipliers, each fraction of (1)

$$= \frac{dx + dy + dz}{(xy - xz) + (yz - yx) + (zx - zy)} = \frac{dx + dy + dz}{0}$$

$$\Rightarrow dx + dy + dz = 0 \quad \text{so that} \quad x + y + z = c_2 \quad \dots (4)$$

From (3) and (4), solution is $\phi(x + y + z, xyz) = 0$, ϕ being an arbitrary function.

Ex. 14. Solve $2y(z - 3)p + (2x - z)q = y(2x - 3)$

Sol. Try yourself.

$$\text{Ans. } \phi(x^2 - 3x - z^2 + 6z, x + y^2 - 2z) = 0$$

Ex. 15. Solve $x^2(\partial z / \partial x) + y^2(\partial z / \partial y) = (x + y)z$. [Delhi B.A. (Prog.) II, 2014]

Sol. Re-writing the given equation $x^2p + y^2q = (x + y)z \quad \dots (1)$

The Lagrange's auxiliary equations for (1) are $\frac{dx}{x^2} = \frac{dy}{y^2} = \frac{dz}{(x + y)z} \quad \dots (2)$

Taking the first two fractions of (2), $(1/x^2)dx - (1/y^2)dy = 0$.

Integrating, $-(1/x) + (1/y) = C_1$ or $1/y - 1/x = C_1 \quad \dots (3)$

Choosing $1/x, 1/y, -1/z$ as multipliers, each fraction of (2)

$$= \frac{(1/x)dx + (1/y)dy - (1/z)dz}{x + y - (x + y)} = \frac{(1/x)dx + (1/y)dy - (1/z)dz}{0}$$

$$\therefore (1/x)dx + (1/y)dy - (1/z)dz = 0 \quad \text{so that} \quad xyz = C_2 \quad \dots (4)$$

From (3) and (4), solution is $\Phi(1/y - 1/x, xyz) = 0$, Φ being an arbitrary function.

EXERCISE 2(C)

Solve the following partial differential equations:

$$1. \quad x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2) \quad \text{Ans. } \phi(x^2 + y^2 + z^2, xyz) = 0$$

[Madras 2012; Mysore 2004, Delhi B.Sc. (Prog.) II 2007, M.S. Univ. T.N. 2007]

$$2. \quad z(xp - yq) = y^2 - x^2 \quad [\text{Delhi B.A. (Prog.) II 2012}] \quad \text{Ans. } \phi(x^2 + y^2 + z^2, xy) = 0$$

$$3. \quad (y^2 + z^2)p - xyq + xz = 0 \quad \text{Ans. } \phi(x^2 + y^2 + z^2, y/z) = 0$$

$$4. \quad yp - xq = 2x - 3y \quad [\text{M.S. Univ. T.N. 2007}] \quad \text{Ans. } \phi(x^2 + y^2, 3x + 2y + z) = 0$$

$$5. \quad x^2(y - z)p + y^2(z - x)q = z^2(x - y) \quad \text{Ans. } \phi(xyz, 1/x + 1/y + 1/z) = 0$$

[Agra 2001; Meerut 2007, 12, Bilaspur 2004, Rewa 2003]

2.11 Type 4 based on Rule IV for solving $(dx)/P = (dy)/Q = (dz)/R$ (1)

Let P_1, Q_1 and R_1 be functions of x, y and z . Then, by a well-known principle of algebra, each fraction of (1) will be equal to $(P_1dx + Q_1dy + R_1dz) / (P_1P + Q_1Q + R_1R)$ (2)

Suppose the numerator of (2) is exact differential of the denominator of (2). Then (2) can be combined with a suitable fraction in (1) to give an integral. However, in some problems, another set of multipliers P_2, Q_2 and R_2 are so chosen that the fraction

$$(P_2dx + Q_2dy + R_2dz) / (P_2P + Q_2Q + R_2R) \quad \dots (3)$$

is such that its numerator is exact differential of denominator. Fractions (2) and (3) are then combined to give an integral. This method may be repeated in some problems to get another integral. Sometimes only one integral is possible by using the above rule IV. In such cases second integral should be obtained by using rule 1 of Art. 2.5 or rule 2 of Art. 2.7 or rule 3 of Art. 2.9.

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2.12 Solved examples based in Art. 2.11

Ex. 1. Solve $(y+z)p + (z+x)q = x+y$. [Indore 2000; Jabalpur 2000, Jiwaji 2002, Agra 2011; Kanpur 2008; Purvanchal 2007, Ravishankar 2005; Rohilkhand 2008]

Sol. Here the Lagrange's auxiliary equations are $\frac{dx}{y+z} = \frac{dy}{z+x} = \frac{dz}{x+y}$ (1)

Choosing 1, -1, 0 as multipliers, each fraction of (1) = $\frac{dx-dy}{(y+z)-(z+x)} = \frac{d(x-y)}{-(x-y)}$ (2)

Again, choosing 0, 1, -1 as multipliers, each fraction of (1) = $\frac{dy-dz}{(z+x)-(x+y)} = \frac{d(y-z)}{-(y-z)}$ (3)

Finally, choosing 1, 1, 1 as multipliers, each fraction of (1) = $\frac{dx+dy+dz}{(y+z)+(z+x)+(x+y)} = \frac{d(x+y+z)}{2(x+y+z)}$ (4)

(2), (3) and (4) $\Rightarrow \frac{d(x-y)}{-(x-y)} = \frac{d(y-z)}{-(y-z)} = \frac{d(x+y+z)}{2(x+y+z)}$ (5)

Taking the first two fractions of (5), $\frac{d(x-y)}{x-y} = \frac{d(y-z)}{y-z}$.

Integrating, $\log(x-y) = \log(y-z) + \log c_1$, c_1 being an arbitrary constant.
or $\log \{(x-y)/(y-z)\} = \log c_1$ or $(x-y)/(y-z) = c_1$ (6)

Taking the first and the third fractions of (5), $2 \frac{d(x-y)}{(x-y)} + \frac{d(x+y+z)}{x+y+z} = 0$

Integrating, $2 \log(x-y) + \log(x+y+z) = \log c_2$ or $(x-y)^2(x+y+z) = c_2$ (7)

From (6) and (7), the required general solution is

$\phi[(x-y)^2(x+y+z), (x-y)/(y-z)] = 0$, ϕ being an arbitrary function.

Ex. 2. Solve $y^2(x-y)p + x^2(y-x)q = z(x^2+y^2)$ [Meerut 2017; Nagpur 2010]

Sol. Here the Lagrange's auxiliary equations for the given equation are

$$\frac{dx}{y^2(x-y)} = \frac{dy}{-x^2(x-y)} = \frac{dz}{z(x^2+y^2)} \quad \dots (1)$$

Taking the first two fractions of (1), $x^2 dx = -y^2 dy$ or $3x^2 dx + 3y^2 dy = 0$.

Integrating, $x^3 + y^3 = c_1$, c_1 being an arbitrary constant. ... (2)

Choosing 1, -1, 0 as multipliers, each fraction of (1)

$$= \frac{dx-dy}{y^2(x-y)+x^2(x-y)} = \frac{dx-dy}{(x-y)(x^2+y^2)} \quad \dots (3)$$

Combining the third fraction of (1) with fraction (3), we get

$$\frac{dx-dy}{(x-y)(x^2+y^2)} = \frac{dz}{z(x^2+y^2)} \quad \text{or} \quad \frac{d(x-y)}{x-y} - \frac{dz}{z} = 0.$$

Integrating, $\log(x-y) - \log z = \log c_2$ or $(x-y)/z = c_2$ (4)

From (3) and (4), solution is $\phi(x^3+y^3, (x-y)/z) = 0$, ϕ being an arbitrary function.

Ex. 3. Solve $(x^2-y^2-z^2)p + 2xyq = 2xz$ or $(y^2+z^2-x^2)p - 2xyq = -2xz$.

[Avadh 2010; Lucknow 2009; Bhopal 2010]

Sol. Here the Lagrange's auxiliary equations for the given equation are

$$\frac{dx}{y^2+z^2-x^2} = \frac{dy}{-2xy} = \frac{dz}{-2xz} \quad \dots(1)$$

Taking the last two fractions of (1), we have

$$(1/y)dy = (1/z)dz \quad \text{so that} \quad (1/y)dy - (1/z)dz = 0.$$

$$\text{Integrating, } \log y - \log z = \log c_1 \quad \text{or} \quad y/z = c_1. \quad \dots(2)$$

Choosing x, y, z as multipliers, each fraction of (1)

$$= \frac{x dx + y dy + z dz}{xy^2 + xz^2 - x^3 - 2xy^2 - 2xz^2} = \frac{x dx + y dy + z dz}{-x(x^2 + y^2 + z^2)} \quad \dots(3)$$

Combining the third fraction of (1) with fraction (3), we have

$$\frac{x dx + y dy + z dz}{-x(x^2 + y^2 + z^2)} = \frac{dz}{-2xz} \quad \text{or} \quad \frac{2x dx + 2y dy + 2z dz}{x^2 + y^2 + z^2} - \frac{dz}{z} = 0.$$

$$\text{Integrating, } \log(x^2 + y^2 + z^2) - \log z = \log c_2 \quad \text{or} \quad (x^2 + y^2 + z^2)/z = c_2. \quad \dots(4)$$

From (2) and (4) solution is $\phi(y/z, (x^2 + y^2 + z^2)/z) = 0$, ϕ being an arbitrary function.

Ex. 4. Solve $(1+y)p + (1+x)q = z$. [M.S. Univ. T.N. 2007; Kanpur 2011]

Sol. Here the Lagrange's auxiliary equations are

$$\frac{dx}{1+y} = \frac{dy}{1+x} = \frac{dz}{z} \quad \dots(1)$$

Taking the first two fractions of (1), we have

$$(1+x)dx = (1+y)dy \quad \text{or} \quad 2(1+x)dx - 2(1+y)dy = 0.$$

$$\text{Integrating, } (1+x)^2 - (1+y)^2 = c_1, \quad c_1 \text{ being an arbitrary constant.} \quad \dots(2)$$

$$\text{Taking } 1, 1, 0 \text{ as multipliers, each fraction of (1)} \quad = \frac{dx+dy}{1+y+1+x} = \frac{d(2+x+y)}{2+x+y} \quad \dots(3)$$

Combining the last fraction of (1) with fraction (3), we get

$$\frac{d(2+x+y)}{2+x+y} = \frac{dz}{z} \quad \text{or} \quad \frac{d(2+x+y)}{2+x+y} - \frac{dz}{z} = 0.$$

$$\text{Integrating, } \log(2+x+y) - \log z = \log c_2 \quad \text{or} \quad (2+x+y)/z = c_2. \quad \dots(4)$$

From (2) and (4), the required general solution is given by

$$\phi[(1+x)^2 - (1+y)^2, (2+x+y)/z] = 0, \quad \phi \text{ being an arbitrary function.}$$

Ex. 5. Find the general integral of $xzp + yzq = xy$.

Sol. Try yourself.

Ans. $\phi(x/y, xy - z^2) = 0$, ϕ being an arbitrary function.

Ex. 6. Solve $(x^2 - yz)p + (y^2 - zx)q = z^2 - xy$. Delhi B.Sc. (Hons.) 2005, 11, M.D.U.

Rohtak 2005; Agra 2008, 09, 12; Guwahati 2007; Meerut 2006; Sagar 2000;

Ravishankar 2000; Lucknow 2010]

Sol. Here the Lagrange's auxiliary equations are

$$\frac{dx}{x^2 - yz} = \frac{dy}{y^2 - zx} = \frac{dz}{z^2 - xy} \quad \dots(1)$$

Choosing 1, -1, 0 and 0, 1, -1 as multipliers in turn, each fraction of (1)

$$= \frac{dx - dy}{x^2 - y^2 + z(x - y)} = \frac{dy - dz}{(y - z)(y + z + x)}$$

$$\text{so that} \quad \frac{dx - dy}{(x - y)(x + y + z)} = \frac{dy - dz}{(y - z)(y + z + x)} \quad \text{or} \quad \frac{d(x - y)}{x - y} - \frac{d(y - z)}{y - z} = 0.$$

$$\text{Integrating, } \log(x - y) - \log(y - z) = \log c_2 \quad \text{or} \quad (x - y)/(y - z) = c_1. \quad \dots(2)$$

Choosing x, y, z as multipliers, each fraction of (1)

$$= \frac{xdx + ydy + zdz}{x^3 + y^3 + z^3 - 3xyz} = \frac{xdx + ydy + zdz}{(x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)} \quad \dots(3)$$

Again, choosing 1, 1, 1 as multipliers, each fraction of (1)

$$= \frac{dx + dy + dz}{x^2 + y^2 + z^2 - xy - yz - zx} \quad \dots(4)$$

From (3) and (4),
$$\frac{xdx + ydy + zdz}{x + y + z} = dx + dy + dz$$

or
$$2(x + y + z) d(x + y + z) - (2xdx + 2ydy + 2zdz) = 0.$$

Integrating,
$$(x + y + z)^2 - (x^2 + y^2 + z^2) = 2c_2$$

or
$$(x^2 + y^2 + z^2 + 2xy + 2yz + 2zx) - (x^2 + y^2 + z^2) = 2c_2$$

or
$$xy + yz + zx = c_2, c_2 \text{ being an arbitrary constant.} \quad \dots(5)$$

From (2) and (5), the required general solution is given by

$$\phi[xy + yz + zx, (x - y)/(y - z)] = 0, \phi \text{ being an arbitrary function.}$$

Ex. 7. Solve $(x^2 - y^2 - yz)p + (x^2 - y^2 - zx)q = z(x - y)$.

Sol. Here Lagrange's auxiliary equations for the given equation are

$$\frac{dx}{x^2 - y^2 - yz} = \frac{dy}{x^2 - y^2 - zx} = \frac{dz}{z(x - y)} \quad \dots(1)$$

Choosing 1, -1, 0 as multipliers, each fraction of (1)

$$= \frac{dx - dy}{(x^2 - y^2 - yz) - (x^2 - y^2 - zx)} = \frac{dx - dy}{z(x - y)} \quad \dots(2)$$

Choosing $x, -y, 0$ as multipliers each fraction of (1)

$$= \frac{xdx - ydy}{x(x^2 - y^2 - yz) - y(x^2 - y^2 - zx)} = \frac{xdx - ydy}{(x - y)(x^2 - y^2)} \quad \dots(3)$$

From (1), (2), (3) we have

$$\frac{dz}{z(x - y)} = \frac{dx - dy}{z(x - y)} = \frac{xdx - ydy}{(x - y)(x^2 - y^2)} \quad \text{or} \quad \frac{dz}{z} = \frac{dx - dy}{z} = \frac{2xdx - 2ydy}{2(x^2 - y^2)} \quad \dots(4)$$

Taking the first two fractions of (4), we have

$$dz = dx - dy \quad \text{so that} \quad z - x + y = c_1 \quad \dots(5)$$

Again, taking the first and third fractions of (4),
$$d(x^2 - y^2)/(x^2 - y^2) - (2/z) dz = 0$$

Integrating, $\log(x^2 - y^2) - 2 \log z = c_2$ or $(x^2 - y^2)/z^2 = c_2 \quad \dots(6)$

From (5) and (6), solution is $\phi(z - x + y, (x^2 + y^2)/z^2) = 0$, ϕ being an arbitrary function.

Ex. 8. Solve $(x^2 + y^2 + yz)p + (x^2 + y^2 - xz)q = z(x + y)$.

Sol. Try yourself.

Ans. $\phi(z - x + y, (x^2 + y^2)/z^2) = 0$

Ex. 9. Solve $\cos(x + y)p + \sin(x + y)q = z$. [Garhwal 2010; Kanpur 2014; Meerut 2007; Delhi B.Sc. (Hons.) 2007; Avadh 2009; Delhi B.A./B.Sc. (Prog.) 2007]

Sol. Here the Lagrange's auxiliary equations are
$$\frac{dx}{\cos(x + y)} = \frac{dy}{\sin(x + y)} = \frac{dz}{z} \quad \dots(1)$$

Choosing 1, 1, 0 as multipliers, each fraction of (1)

$$= \frac{dx + dy}{\cos(x + y) + \sin(x + y)} = \frac{d(x + y)}{\cos(x + y) + \sin(x + y)} \quad \dots(2)$$

Choosing 1, -1, 0 as multipliers, each fraction of (1) $= \frac{dx-dy}{\cos(x+y)-\sin(x+y)} \dots(3)$

From (1), (2) and (3), $\frac{dz}{z} = \frac{d(x+y)}{\cos(x+y)+\sin(x+y)} = \frac{dx-dy}{\cos(x+y)-\sin(x+y)} \dots(4)$

Taking the first two fractions of (4), $\frac{dz}{z} = \frac{d(x+y)}{\cos(x+y)+\sin(x+y)} \dots(5)$

Putting $x+y=t$ so that $d(x+y)=dt$, (5) reduces to

$$\frac{dz}{z} = \frac{dt}{\cos t + \sin t} = \frac{dt}{\sqrt{2} \left\{ \left(\frac{1}{\sqrt{2}} \right) \cos t + \left(\frac{1}{\sqrt{2}} \right) \sin t \right\}} = \frac{dt}{\sqrt{2} \{ \sin(\pi/4) \cos t + \cos(\pi/4) \sin t \}} = \frac{dt}{\sqrt{2} \sin(t + \pi/4)}$$

Thus, $(\sqrt{2}/z)dz = \operatorname{cosec}(t + \pi/4) dt$.

Integrating, $\sqrt{2} \log z = \log \tan \frac{1}{2} \left(t + \frac{\pi}{4} \right) + \log c_1$, or $z^{\sqrt{2}} = c_1 \tan \left(\frac{t}{2} + \frac{\pi}{8} \right)$

or $z^{\sqrt{2}} \cot \left(\frac{x+y}{2} + \frac{\pi}{8} \right) = c_1$, as $t = x+y$ $\dots(6)$

Taking the last two fraction of (4), $dx-dy = \frac{\cos(x+y)-\sin(x+y)}{\cos(x+y)+\sin(x+y)} d(x+y)$. $\dots(7)$

On R.H.S. of (7), putting $x+y=t$, so that $d(x+y)=dt$, (7) reduces to

$$dx-dy = \frac{\cos t - \sin t}{\cos t + \sin t} dt. \quad \text{so that} \quad x-y = \log(\sin t + \cos t) - \log c_2$$

or $(\sin t + \cos t)/c_2 = e^{x-y}$ or $e^{(x-y)}(\sin t + \cos t) = c_2$
or $e^{y-x}[\sin(x+y) + \cos(x+y)] = c_2$, as $t = x+y$. $\dots(8)$

From (6) and (8), the required general solution is

$$\phi \left[z^{\sqrt{2}} \cot \left(\frac{x+y}{2} + \frac{\pi}{8} \right), e^{y-x} \{ \sin(x+y) + \cos(x+y) \} \right] = 0, \text{ where } \phi \text{ is an arbitrary function.}$$

Ex. 10. Solve $\cos(x+y)p + \sin(x+y)q = z + (1/z)$. [Delhi B.A. (Prog.) 2011]

Sol. Do like Ex. 9. **Ans.** $\phi \left[(z^2+1)^{1/\sqrt{2}} \tan \left(\frac{3\pi}{8} - \frac{x+y}{2} \right), e^{y-x} \{ \cos(x+y) + \sin(x+y) \} \right] = 0$

Ex. 11. Solve $xp + yq = z - a \sqrt{(x^2 + y^2 + z^2)}$. [Rohilkhand 2004]

Sol. Try yourself.

$$\text{Ans. } \phi[x^{a-1}/\{z + \sqrt{(x^2 + y^2 + z^2)}\}, x/y] = 0$$

Ex. 12. Solve $p+q=x+y+z$. [Bhopal 2010, Kanpur 2010; Meerut 2013; Gulberge 2005]

Sol. Here Lagrange's auxiliary equations are $\frac{dx}{1} = \frac{dy}{1} = \frac{dz}{x+y+z}$. $\dots(1)$

Taking the first two fractions of (1), $dx-dy=0$ so that $x-y=c_1$. $\dots(2)$

Choosing 1, 1, 1 as multipliers, each fraction of (1) $= \frac{dx+dy+dz}{1+1+(x+y+z)} = \frac{d(2+x+y+z)}{2+x+y+z}$ $\dots(3)$

Combining the first fraction of (1) with fraction (3), $d(2+x+y+z)/(2+x+y+z) = dx$.

Integrating, $\log(2+x+y+z) - \log c_2 = x$ or $(2+x+y+z)/c_2 = e^x$
or $e^{-x}(2+x+y+z) = c_2$, c_2 being arbitrary function $\dots(4)$

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From (2) and (4), the required general solution is

$$\phi[x - y, e^x(2 + x + y + z)] = 0, \phi \text{ being an arbitrary function.}$$

Ex. 13. Solve $(2x^2 + y^2 + z^2 - 2yz - zx - xy)p + (x^2 + 2y^2 + z^2 - yz - 2zx - xy)q = x^2 + y^2 + 2z^2 - yz - zx - 2xy$.

Sol. Here Lagrange's auxiliary equations are

$$\frac{dx}{2x^2 + y^2 + z^2 - 2yz - zx - xy} = \frac{dy}{x^2 + 2y^2 + z^2 - yz - 2zx - xy} = \frac{dz}{x^2 + y^2 + 2z^2 - yz - zx - 2xy} \dots (1)$$

Choosing 1, -1, 0; 0, 1, -1 and -1, 0, 1 as multipliers in turn, each fraction of (1)

$$\begin{aligned} &= \frac{dx - dy}{x^2 - y^2 - yz + zx} = \frac{dy - dz}{y^2 - z^2 - zx + xy} = \frac{dz - dx}{z^2 - x^2 - xy + yz} \\ \therefore &\frac{dx - dy}{(x - y)(x + y + z)} = \frac{dy - dz}{(y - z)(x + y + z)} = \frac{dz - dx}{(z - x)(x + y + z)} \dots (2) \end{aligned}$$

Taking the first two fractions of (2), we have

$$(dx - dy)/(x - y) - (dy - dz)/(y - z) = 0.$$

$$\text{Integrating, } \log(x - y) - \log(y - z) = \log c_1 \quad \text{or} \quad (x - y)/(y - z) = c_1 \dots (3)$$

Taking the last two fractions of (2),

$$(dy - dz)/(y - z) - (dz - dx)/(z - x) = 0.$$

$$\text{Integrating, } \log(y - z) - \log(z - x) = \log c_2 \quad \text{or} \quad (y - z)/(z - x) = c_2 \dots (4)$$

From (3) and (4), the required general solution is

$$\phi[(x - y)/(y - z), (y - z)/(z - x)] = 0, \phi \text{ being an arbitrary function.}$$

Ex. 14. Find the general solution of the partial differential equation $px(x + y) - qy(x + y) + (x - y)(2x + 2y + z) = 0$. [Delhi B.A. II (Prog) 2009, 12, 14; Delhi B.Sc. Hons. 2006, 09, 11]

Sol. Given

$$x(x + y)p - y(x + y)q = -(x - y)(2x + 2y + z) \dots (1)$$

$$\text{Lagrange's auxiliary equations are } \frac{dx}{x(x + y)} = \frac{dy}{-y(x + y)} = \frac{dz}{-(x - y)(2x + 2y + z)} \dots (2)$$

$$\text{Taking the first two fractions, } (1/x)dx = -(1/y)dy \quad \text{or} \quad (1/x)dx + (1/y)dy = 0.$$

$$\text{Integrating, } \log x + \log y = \log c_1 \quad \text{or} \quad xy = c_1 \dots (3)$$

Again, each fraction of (2)

$$\begin{aligned} &= \frac{dx + dy}{x(x + y) - y(x + y)} = \frac{dx + dy + dz}{x(x + y) - y(x + y) - (x - y)(2x + 2y + z)} \\ &= \frac{dx + dy}{(x - y)(x + y)} = \frac{dx + dy + dz}{(x - y)(x + y) - (x - y)(2x + 2y + z)} \end{aligned}$$

$$\text{Thus, } \frac{dx + dy}{(x + y)} = \frac{dx + dy + dz}{x + y - (2x + 2y + z)} = -\frac{dx + dy + dz}{x + y + z}$$

$$\text{Thus, } \frac{dx + dy}{x + y} + \frac{dx + dy + dz}{x + y + z} = 0, \quad \text{so that } \log(x + y) + \log(x + y + z) = \log c_2$$

$$\text{or} \quad (x + y)(x + y + z) = c_2, c_2 \text{ being an arbitrary constant.} \dots (4)$$

From (3) and (4), solution is $\phi[xy, (x + y)(x + y + z)] = 0$, ϕ being an arbitrary function.

Ex. 15. Solve $px(z - 2y^2) = (z - qy)(z - y^2 - 2x^2)$. [Delhi B.Sc. (Hons.) 2002]

$$\text{Sol. Re-writing the given equation } x(z - 2y^2)p + y(z - y^2 - 2x^2)q = z(z - y^2 - 2x^2) \dots (1)$$

Lagrange's auxiliary equations for (1) are $\frac{dx}{x(z-2y^2)} = \frac{dy}{y(z-y^2-2x^2)} = \frac{dz}{z(z-y^2-2x^2)}$... (2)

Taking the last two fractions, $(1/y) dy - (1/z) dz = 0$ so that $y/z = C_1$... (3)

Taking 0, $-2y$, 1 as multipliers, each fraction of (2)

$$-\frac{-2ydy + dz}{-2y^2(z-y^2-2x^2) + z(z-y^2-2x^2)} = \frac{d(z-y^2)}{(z-2y^2)(z-y^2-2x^2)} \quad \dots (4)$$

Combining fraction (4) with first fraction of (2), we get

$$\frac{dx}{x(z-2y^2)} = \frac{d(z-y^2)}{(z-2y^2)(z-y^2-2x^2)} \quad \text{or} \quad \frac{d(z-y^2)}{dx} = \frac{z-y^2-2x^2}{x}$$

$$\text{or} \quad du/dx = (u-2x^2)/x, \text{ taking } z-y^2 = u \quad \dots (5)$$

or $(du/dx) - (1/x)u = -2x$ which is an ordinary linear differential equation

whose I.F. = $e^{-\int(1/x)dx} = e^{-\log x} = e^{\log x^{-1}} = x^{-1} = 1/x$ and solution is

$$u \cdot \frac{1}{x} = \int(-2x) \left(\frac{1}{x}\right) dx + C_2 \quad \text{or} \quad \frac{z-y^2}{x} = -2x + C_2, \text{ using (5)}$$

$$\text{or } (z-y^2)/x + 2x = C_2 \quad \text{or} \quad (z-y^2+2x^2)/x = C_2 \quad \dots (6)$$

From (3) and (6), the required general solution of (1)

$$\Phi(y/z, (z-y^2-2x^2)/x) = 0, \Phi \text{ being an arbitrary function.}$$

Ex. 16. Solve $px(z-2y^2) = (z-xy)(z-y^2-2x^3)$.

Sol. Do like Ex. 15,

$$\text{Ans. } \Phi(y/z, (z-y^2+x^3)/x) = 0$$

Ex. 17. Solve $2x(y+z^2)p + y(2y+z^2)q = z^3$

[Delhi B.Sc. (Hons.) 2007]

Sol. The Lagrange's auxiliary equations for the given equation are

$$\frac{dx}{2x(y+z^2)} = \frac{dy}{y(2y+z^2)} = \frac{dz}{z^3} \quad \dots (1)$$

$$\text{Each fraction of (1)} = \frac{dx}{2x(y+z^2)} = \frac{z dy + y dz}{2yz(y+z^2)} = \frac{d(yz)}{2yz(y+z^2)}$$

$$\therefore (1/x) dx + (1/yz) d(yz) = 0 \quad \text{so that} \quad x/(yz) = C_1 \quad \dots (2)$$

$$\text{From the last two fractions of (1), } \frac{dy}{dz} = \frac{y(2y+z^2)}{z^3} = \frac{2y^2}{z^3} + \frac{y}{z} \text{ or } y^{-2} \frac{dy}{dz} - \frac{1}{z} y^{-1} = \frac{2}{z^3} \quad \dots (3)$$

Putting $-y^{-1} = u$ and $(1/y^2) \times (dy/dz) = du/dz$ in (3), we get

$$(du/dz) + (1/z)u = 2/z^3, \text{ which is an ordinary linear equation.}$$

$$\text{Its I.F.} = e^{\int(1/z)dz} = e^{\log z} = z \text{ and solution is } uz = \int(2/z^3) z dz - C_2 = -2z^{-1} - C_2$$

$$\text{or} \quad -y^{-1}z - 2z^{-1} = -C_2 \quad \text{or} \quad z/y - 2/z = C_2 \quad \dots (4)$$

From (3) and (4), solution is $\Phi(x/yz, z/y - 2/z) = 0$, ϕ being arbitrary function.

2.18 Ordinary and Partial Differential Equations

Ex. 18. $xp + zq + y = 0$.

[M.D.U. Rohtak 2004]

Sol. Try yourself.

Ans. $\log |x| - \tan^{-1}(y/z) = \phi(y^2 + z^2)$

Ex. 19. Find the general solution of the differential equation $x^2(\partial z/\partial x) + y^2(\partial z/\partial y) = (x+y)z$. [Delhi B.Sc. (Hons.) III 2012; Delhi B.A./B.Sc. (Prog.) II 2007, 16]

Sol. Let $p = \partial z/\partial x$ and $q = \partial z/\partial y$. Then, the given equation takes the form

$$x^2p = y^2q = z(x+y) \quad \dots (1)$$

The Lagrange's auxiliary equations for (1) are

$$(dx)/x^2 = (dy)/y^2 = (dz)/z(x+y) \quad \dots (2)$$

Taking the first two fractions of (2), $(1/x^2)dx - (1/y^2)dy = 0$

$$\text{Integrating, } -(1/x) + (1/y) = c_1 \quad \text{or} \quad (x-y)/xy = c_1 \quad \dots (3)$$

$$\text{Choosing } 1, -1, 0 \text{ as multipliers, each fraction of (2) } = \frac{dx-dy}{x^2-y^2} \quad \dots (4)$$

Combining the last fraction of (2) with fraction (4), we have

$$\frac{dx-dy}{(x-y)(x+y)} = \frac{dz}{z(x+y)} \quad \text{or} \quad \frac{dx-dy}{x-y} - \frac{dz}{z} = 0$$

$$\text{Integrating, } \log(x-y) - \log z = \log c_2 \quad \text{or} \quad (x-y)/z = c_2 \quad \dots (5)$$

$$\text{From (5), } x-y = c_2 z \quad \dots (6)$$

$$\text{using (6), (3) becomes } (c_2 z)/xy = a \quad \text{or} \quad (xy)/z = c_2/c_1 = c_3 \text{ say } \dots (7)$$

From (5) and (7), the required solution is $\phi((x,y)/z, (x-y)/z) = 0$.

EXERCISE 2(D)

Solve the following partial differential equations:

$$1. (x^2 + y^2)p + 2xy q = z(x+y) \quad \text{Ans. } (x+y)/z = \phi(y/(x^2 - y^2))$$

$$2. \{y(x+y) + az\}p + \{x(x+y) - az\}q = z(x+y) \quad \text{Ans. } (x+y)/z = \phi(x^2 - y^2 - 2az)$$

$$3. (y^2 + yz + z^2)p + (z^2 + zx + x^2)q = x^2 + xy + y^2 \quad \text{Ans. } \phi\left(\frac{y-z}{x-y}, \frac{x-z}{x-y}\right) = 0$$

2.13 Miscellaneous example on $Pp + Qq = R$

Example. (a). Find the surface whose tangent planes cut off an intercept of constant length k from the axis of z .

(b) Formulate partial differential equation for surfaces whose tangent planes form a tetrahedron of constant volume with the coordinate planes.

Sol. (a) We know that the equation of the tangent plane at point (x, y, z) to a surface is given by

$$p(X-x) + q(Y-y) = Z-z, \quad \dots (1)$$

where X, Y, Z denote current coordinates of any point on the plane (1). Since (1) cuts an intercept k on the z -axis, it follows that (1) must pass through the point $(0, 0, k)$. Hence putting $X=0, Y=0$ and $Z=k$ in (1), we obtain

$$px + qy = z - k, \quad \dots (2)$$

which is well known Lagrange's linear equation. For (2), the Lagrange's auxiliary equations are

$$(dx)/x = (dy)/y = (dz)/(z-x). \quad \dots (3)$$

$$\text{Taking the first two fractions of (3), } (1/x)dx - (1/y)dy = 0. \quad \text{so that } x/y = c_1. \quad \dots (4)$$